

Odisha Tourism Smart Assistant Chatbot

AI-Powered Multilingual Tourism Guidance System for Odisha

Abstract

The Odisha Tourism Smart Assistant Chatbot is an AI-powered multilingual tourism guidance platform designed to assist tourists visiting Odisha from different regions of India and abroad. Many tourists face challenges related to transportation, travel planning, route guidance, local culture, food information, budget estimation, and nearby attraction discovery due to lack of prior knowledge about Odisha. Existing tourism websites provide static information, but users often require quick conversational assistance.

This project introduces an intelligent chatbot system integrated with Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), vector databases, and map services to provide instant, accurate, and context-aware responses to tourist queries. The chatbot supports multiple languages including English, Hindi, and Odia to ensure accessibility for a broad user base.

The system is capable of answering questions regarding tourist destinations, route planning, travel time, budget estimation, nearby attractions, local food, culture, festivals, and transportation facilities. The chatbot integrates Google Maps services for route and navigation support while using ChromaDB for semantic retrieval of tourism-related knowledge.

The proposed architecture combines modern AI technologies such as Gemini 3.1 Pro or OpenAI models, embedding models, ChromaDB vector storage, FastAPI backend services, and multilingual processing modules. The system is scalable and capable of handling peak tourist traffic efficiently.

This project aims to enhance tourism accessibility, improve tourist experience, reduce confusion during travel planning, and digitally strengthen Odisha tourism services.

1. Introduction

Tourism is one of the most important sectors contributing to economic growth and cultural development. Odisha is a state rich in culture, heritage, temples, beaches, wildlife, handicrafts, festivals, and traditional food. Every year, thousands of tourists visit famous destinations such as Puri, Konark, Bhubaneswar, Chilika Lake, Simlipal, and Jagannath Temple.

Despite the availability of tourism websites and informational portals, tourists often encounter several problems such as:

- Lack of local travel knowledge
- Difficulty understanding routes and transportation
- Language barriers
- Uncertainty about travel budgets

- Lack of awareness of nearby attractions
- Difficulty accessing quick information during travel
- Confusion regarding local customs and cultural practices

Traditional tourism websites mainly provide static information that may not efficiently solve dynamic user queries. Therefore, there is a need for an intelligent conversational system capable of understanding user questions naturally and providing accurate, contextual, and multilingual responses.

The Odisha Tourism Smart Assistant Chatbot is designed to solve these problems using Artificial Intelligence, Natural Language Processing, Retrieval-Augmented Generation, and vector-based semantic search.

2. Problem Statement

Tourists visiting Odisha often face difficulties in obtaining real-time and contextual travel information. Existing tourism websites provide limited static content and lack conversational support for dynamic queries.

Users frequently require information such as:

- Distance between locations
- Route directions
- Estimated travel budgets
- Nearby tourist attractions
- Local food recommendations
- Cultural guidance
- Temple rules and timings
- Transportation facilities
- Suggested travel plans

Additionally, language barriers create difficulties for tourists who are not comfortable with English.

Therefore, there is a need for a multilingual AI-powered chatbot capable of answering tourism-related questions intelligently and accurately.

3. Objectives

The main objectives of this project are:

1. To develop a multilingual tourism chatbot for Odisha.
2. To provide instant answers to tourism-related queries.
3. To support English, Hindi, and Odia languages.
4. To provide route guidance using map services.
5. To estimate travel budgets based on user preferences.
6. To suggest nearby tourist attractions.
7. To provide information about food, culture, festivals, and traditions.
8. To improve the overall tourism experience.

9. To integrate AI-based Retrieval-Augmented Generation for accurate responses.
 10. To build a scalable architecture capable of handling large tourist traffic.
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4. Scope of the Project

The scope of the Odisha Tourism Smart Assistant Chatbot includes:

- Tourism guidance
- Intelligent conversational support
- Route planning and navigation assistance
- Multilingual communication
- Budget estimation
- Cultural and food recommendations
- Nearby attraction recommendations
- Tourism knowledge retrieval
- AI-powered semantic search
- Mobile and web accessibility

The system can be integrated into:

- Odisha tourism websites
 - Mobile applications
 - Tourism kiosks
 - Smart tourism platforms
 - Hotel and travel booking systems
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5. Proposed System

The proposed system is an AI-powered multilingual chatbot integrated with modern AI technologies and tourism databases.

The chatbot will:

- Understand user queries using Natural Language Processing.
- Retrieve tourism-related information using ChromaDB.
- Generate contextual responses using Gemini 3.1 Pro or OpenAI models.
- Support multilingual interaction.
- Provide route guidance using Google Maps APIs.
- Suggest nearby tourist destinations.
- Estimate travel costs dynamically.

The system combines structured tourism databases and semantic retrieval systems to ensure accurate and reliable responses.

6. Key Features

6.1 Multilingual Support

The chatbot supports:

- English
- Hindi
- Odia

Users can communicate naturally in their preferred language.

6.2 Tourism Question Answering

The chatbot answers tourism-related questions such as:

- Best places to visit
 - Famous temples
 - Beach information
 - Food recommendations
 - Cultural details
 - Travel guidance
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6.3 Distance and Route Guidance

The chatbot can:

- Calculate distance between locations
- Estimate travel time
- Generate Google Maps navigation links
- Suggest best routes

Example:

User: "Distance from Bhubaneswar to Puri?"

Response:

- Distance
 - Estimated travel time
 - Navigation link
-

6.4 Budget Estimation

The chatbot estimates travel budgets based on:

- Starting location
- Number of travelers
- Duration of stay
- Travel mode
- Hotel preference

Budget categories:

- Budget
 - Standard
 - Premium
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6.5 Nearby Place Recommendation

The chatbot suggests nearby attractions based on travel routes.

Example:

Route: Bhubaneswar to Puri

Suggested places:

- Dhauli
 - Pipili
 - Konark
 - Raghurajpur
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6.6 Cultural and Food Guidance

The chatbot provides information about:

- Traditional Odia food
 - Festivals
 - Temple customs
 - Local traditions
 - Dress codes
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7. System Architecture

The system architecture consists of multiple layers working together to provide intelligent tourism assistance.

Architecture Components

1. Frontend User Interface
 2. Backend API Layer
 3. AI Processing Layer
 4. Retrieval-Augmented Generation (RAG)
 5. ChromaDB Vector Database
 6. Structured Tourism Database
 7. Google Maps Integration
 8. Translation and Language Processing Module
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8. Technical Overview

8.1 Frontend Layer

The frontend provides the user interaction interface.

Technologies:

- React.js
- Next.js
- Tailwind CSS
- JavaScript

Features:

- Chat bubble UI
 - Responsive design
 - Language selector
 - Quick question buttons
 - Mobile compatibility
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8.2 Backend Layer

The backend handles:

- API requests
- Chat processing
- AI orchestration
- Database communication
- Route services

Technologies:

- FastAPI
- Python

Advantages:

- High performance
 - Asynchronous support
 - Scalable architecture
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8.3 Large Language Model (LLM)

The system uses:

- Gemini 3.1 Pro or
- OpenAI LLMs

Responsibilities:

- Natural language understanding
 - Response generation
 - Multilingual conversation
 - Context-aware communication
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8.4 ChromaDB Vector Database

ChromaDB is used for storing tourism embeddings.

Stored Data:

- Tourist place descriptions
- FAQs
- Travel guides
- Cultural information
- Food recommendations

Advantages:

- Fast semantic retrieval
 - Lightweight architecture
 - Efficient vector search
 - RAG integration support
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8.5 Embedding Models

Embedding models convert text into numerical vectors.

Recommended Models:

- text-embedding-004
 - text-embedding-3-small
 - text-embedding-3-large
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8.6 Structured Database

Structured databases store:

- User data
- Chat history
- Route information
- Hotel details
- Budget tables

Database Options:

- PostgreSQL
 - MySQL
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8.7 Maps and Navigation Integration

Google Maps API is used for:

- Route calculation
 - Distance estimation
 - Navigation support
 - Travel time calculation
 - Nearby attraction identification
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9. Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation is the core intelligence mechanism of the chatbot.

Working Process:

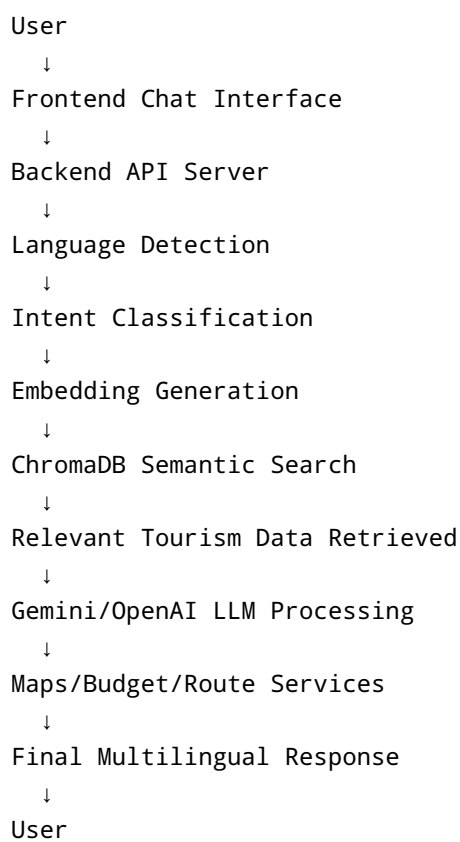
1. User submits query.
2. Query converted into embeddings.
3. ChromaDB searches relevant tourism documents.
4. Retrieved documents are passed to the LLM.
5. LLM generates contextual response.
6. Final response returned to user.

Benefits:

- Improved response accuracy
 - Reduced hallucination
 - Tourism-specific answers
 - Context-aware responses
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10. Flow Diagram

Overall System Flow



11. Functional Modules

11.1 User Interface Module

Responsible for:

- Chat interaction
- Language selection
- User experience

11.2 Query Processing Module

Responsible for:

- Intent detection
- Entity extraction
- Query classification

11.3 Retrieval Module

Responsible for:

- Embedding search
- Semantic retrieval
- Context extraction

11.4 Response Generation Module

Responsible for:

- AI-generated responses
- Contextual conversation
- Multilingual output

11.5 Budget Estimation Module

Responsible for:

- Travel cost estimation
- Hotel budget calculation
- Food cost estimation

11.6 Route Recommendation Module

Responsible for:

- Route generation
 - Nearby places recommendation
 - Travel time estimation
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12. Sample Use Cases

Use Case 1: Distance Query

User: "How far is Puri from Bhubaneswar?"

System Response:

- Distance
 - Estimated time
 - Google Maps direction
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Use Case 2: Budget Query

User: "What is the minimum budget for a 2-day Puri trip?"

System Process:

- Ask source location
 - Ask number of travelers
 - Estimate total cost
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Use Case 3: Nearby Places Query

User: "What places can I visit on the way to Puri?"

System Response:

- Dhauli
 - Pipili
 - Konark
 - Raghurajpur
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13. Database Design

Structured Database Tables

Users Table

- user_id
- name
- language
- location

Tourist Places Table

- place_id
- place_name
- district
- description
- entry_fee
- opening_time

Budget Table

- transport_cost
- hotel_cost
- food_cost

Chat History Table

- chat_id
 - user_query
 - response
 - timestamp
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14. Non-Functional Requirements

Performance

- Fast response generation
- Low latency

Scalability

- Supports 1000+ users during peak season

Reliability

- Stable response system
- Error handling

Security

- API protection
- Rate limiting
- Data privacy

Availability

- 24/7 access
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15. Deployment Architecture

Deployment Components

- Frontend Web Application
- Backend FastAPI Server
- ChromaDB Vector Store
- PostgreSQL Database
- Google Maps APIs
- Cloud Hosting Infrastructure

Recommended Platforms

- AWS
 - Google Cloud Platform
 - Azure
 - Railway
 - Render
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16. Scalability Planning

Expected Traffic:

- Normal days: 100–500 users
- Peak season: 1000+ users

Optimization Strategies:

- Redis caching
 - Async API processing
 - Query optimization
 - Embedding caching
 - Load balancing
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17. Future Enhancements

Future improvements may include:

- Voice-based chatbot
- AI trip planner
- Personalized recommendations
- Image-based tourist place identification
- Offline support
- Live festival updates
- Hotel booking integration

- Emergency tourism support
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18. Advantages of the System

1. Multilingual accessibility
 2. Intelligent tourism guidance
 3. Accurate route information
 4. Context-aware answers
 5. Improved tourist experience
 6. Faster information access
 7. Tourism promotion support
 8. Scalable architecture
 9. AI-powered recommendation system
 10. Better engagement with tourists
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19. Limitations

1. Requires internet connectivity
 2. Dependency on external APIs
 3. Dynamic tourism data requires regular updates
 4. LLM operational costs may increase with traffic
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20. Conclusion

The Odisha Tourism Smart Assistant Chatbot is an advanced AI-powered multilingual tourism guidance system designed to improve the tourism experience in Odisha. By integrating modern AI technologies such as Large Language Models, ChromaDB vector search, Retrieval-Augmented Generation, multilingual processing, and Google Maps services, the system provides accurate, contextual, and conversational assistance to tourists.

The chatbot addresses common travel challenges including route planning, budget estimation, nearby place recommendations, cultural understanding, and quick tourism information access. The multilingual capabilities ensure accessibility for users from different linguistic backgrounds.

The proposed architecture is scalable, efficient, and suitable for real-world tourism applications. This project has the potential to become a smart tourism platform that can digitally strengthen Odisha tourism services and improve visitor satisfaction.

21. Technology Stack Summary

Component	Technology
Frontend	React.js / Next.js
Backend	FastAPI
Programming Language	Python
LLM	Gemini 3.1 Pro / OpenAI
Vector Database	ChromaDB
Structured Database	PostgreSQL / MySQL
Embeddings	text-embedding-004 / text-embedding-3
Maps Service	Google Maps API
Hosting	AWS / GCP / Azure
Caching	Redis

22. Folder Structure

```
odisha-tourism-chatbot/  
├── backend/  
│   ├── api/  
│   ├── services/  
│   ├── rag/  
│   ├── embeddings/  
│   ├── routes/  
│   ├── database/  
│   ├── models/  
│   └── main.py  
├── frontend/  
│   ├── components/  
│   ├── pages/  
│   ├── services/  
│   └── public/  
├── chromadb/  
├── datasets/  
├── prompts/  
├── docs/  
└── docker/
```

23. Final Project Summary

The Odisha Tourism Smart Assistant Chatbot is a multilingual AI-based tourism support platform developed to provide intelligent and conversational guidance to tourists visiting Odisha. The system combines Large Language Models, Retrieval-Augmented Generation, semantic search using ChromaDB, multilingual processing, and route intelligence to deliver accurate and user-friendly tourism assistance.

The chatbot aims to digitally modernize tourism support services, improve visitor satisfaction, and enhance the accessibility of tourism information across Odisha.